

OPERATING INSTRUCTIONS

GMS800 Series Extractive Gas Analyzers



Basic information for all products
in the series



Document Information

Described Product

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Variants: All device versions

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Glossary

AC	Alternating Current
ATEX	Atmosphères Explosifs: Abbreviation for European Directives that govern safety in potentially explosive atmospheres
CAN	Field bus (Control Area Network) with high data security; especially suitable for safety-relevant applications.
CANopen	Communication protocol for the CAN bus. Standardized as European standard EN 50325-4. (www.can-cia.org).
Ethernet	Cable-based network technology for data networks. Basis for network protocols (e.g. TCP/IP).
IFC	Interference Filter Correlation.
LED	Light emitting diode (small indicator lamp)
PC	Personal Computer.
RS485	Standard for digital serial interfaces.
SELV	Safety/Separated Extra-Low Voltage
SOPAS	SICK Open Portal for Applications and Systems: Family of computer programs to set parameters, capture and calculate data.
SOPAS ET	SOPAS Engineering Tool: PC application program to configure modular system components.
UV	Ultraviolet (ultraviolet light).
UVRAS	UV Resonance Absorption Spectrometry.

Warning Symbols



Hazard (general)



Hazard by corrosive substances



Hazard by voltage



Hazard in potentially explosive atmospheres



Hazard by explosive substances/mixtures



Hazard by toxic substances



Hazard for environment/nature/organisms

Information Symbols



Important technical information for this product



Important information on electric or electronic functions



Nice to know



Supplementary information



Link to information at another place

Warning Levels / Signal Words

WARNING

Risk or hazardous situation which *could* result in severe personal injury or death.

CAUTION

Hazard or unsafe practice which *could* result in personal injury or property damage.

NOTICE

Hazard which *could* result in property damage.

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GMS800

1 Important Information

Main safety information

Main operating information

Intended use

Responsibility of user

1.1

Main hazards**Health risks through dangerous sample gases****WARNING: Danger to life/health risks from gas leaks**

When the sample gas contains noxious substances: Gas dangerous to health can escape when gas leaks occur.

- ▶ Take suitable safety precautions to protect persons.
- ▶ *If leaks are suspected: Carry out a leak tightness check (→ p. 52, §8.4).*

**WARNING: Danger to life/health risks during maintenance work**

When the sample gas contains noxious substances: Escaping sample gas can be an acute danger for persons.

Before opening gas paths (e.g. to clean the filter):

- ▶ Flush gas paths with a neutral gas until the dangerous gases have been completely purged.
- ▶ Take breathing protection precautions as necessary for safety.

Operating safety**WARNING: Hazards by explosive or combustible gases**

- ▶ Do not use the gas analyzer
 - to measure explosive or combustible gases/gas mixtures
 - to measure gases/gas mixtures that can build an explosive gas mixture together with air.

Apart from when the device version is specified for such use.

**NOTICE:**

- ▶ Do not allow any liquids or particles to penetrate the gas analyzer.
- ▶ Prevent condensation in the sample gas path of the gas analyzer.

Operating in potentially explosive atmospheres**WARNING: Explosion hazard in potentially explosive atmospheres**

- ▶ Only use the gas analyzer in potentially explosive atmospheres when the device version is specified for this purpose.

**WARNING: Risk of explosion when disregarding the operating conditions**

When the GMS800 is operated with enclosure purging or enclosure pressurization:

- ▶ Observe the stipulated start-up procedure.^[1]
- ▶ Maintain the stipulated operating conditions.^[1]
- ▶ Do not open the enclosure during operation.

[1] → Supplementary Operating Instructions for Enclosure.

1.2

Main operating information**Start-up:**

- ▶ Ensure leak tightness of gas paths (e.g. filters, valves).
If leaks are suspected: Carry out a leak tightness check (→ p. 52, §8.4).
- ▶ Prevent condensation in the sample gas path of the gas analyzer.
- ▶ Make an adjustment after each start-up (→ p. 37, §6).

Additionally in potentially explosive atmospheres:

- ▶ Make sure the enclosure is tightly closed.
- ▶ *When the GMS800 is fitted with enclosure purging or enclosure pressurization:* Pre-purge the enclosure when stipulated in the device specifications (→ Supplementary Operating Instructions for Enclosure or Operating Instructions for Enclosure Pressurization System).

Operating state:

- ▶ Pay attention to status and malfunction fields (→ Operating Instructions for Control Unit).
- ▶ Make adjustments at regular intervals (→ p. 37, §6).

If an “Alarm” message is displayed:

- ▶ Check the current measured values. Consider the situation.
- ▶ Perform the measures planned at your site for the particular situation.
- ▶ When necessary: Switch the alarm message off (“acknowledge”).

In hazardous situations:

- ▶ Switch-off the emergency switch or main switch of the host system.

Shutting down:

- ▶ *Before shutting down:* Purge the sample gas path with a dry neutral gas to prevent condensation in the measuring system.

1.3

General safety information**Sensitive electronics**

Before signal connections are established (also with plug connections):

- ▶ Disconnect the GMS800 and all connected devices from the mains (switch off).
Otherwise the internal electronics could be damaged.

Hazards during maintenance work

- ▶ *If it is necessary to open the device for setting or repair:* Disconnect the device from all power sources before starting work.
- ▶ *If the open device must be live during work:* This work must be performed by skilled persons who are familiar with potential hazards. If it is necessary to remove or open internal components, live parts could be exposed.
- ▶ Never interrupt protective conductor connections.

Hazards due to unsafe state

- ▶ *If severe damage is visible on or in the device:* Shut device down and secure against unauthorized start-up.
- ▶ *If liquids or particles have penetrated the enclosure:* Shut device down immediately and disconnect mains voltage at external source.

1.4 Intended use

1.4.1 Purpose of the device

GMS800 Series gas analyzers measure simultaneously the concentration of one or several gases in a gas mixture.

Sample gas is extracted at a sampling point and flows from there through the internal measuring system of the gas analyzer (extractive gas analysis principle).

1.4.2 Installation site

- ▶ Only use the GMS800 indoors.
- ▶ Do not use the GMS800
 - in potentially explosive atmospheres
 - to measure combustible or explosive gases
 unless the device version is specified for such use or additional safety precautions are taken.

1.4.3 Application limitations

Sample gas properties

- ▶ Do not feed sample gas to the GMS800
 - which contains substances that can chemically attack components carrying sample gas
 - which contains particles that could deposit in the measuring system
 - which contains gas components that could condense in the measuring system.

Combustible sample gases

A defect in an internal gas path (leak) can create an explosion risk when the GMS800 is used to measure combustible gases or gases that can create a flammable gas mixture when mixed with air. *For such applications:*

- ▶ Check whether the device version is suitable for the application (check the manufacturer's specifications).
- ▶ Check which regulations and laws are valid for the installation site in this case.
- ▶ Check whether additional, suitable safety devices must be installed (e.g. pressurization and enclosure purging with inert gas).

Adverse effects on physical measuring properties

In some applications, certain gas components could interfere with measurement – e.g. because they create a similar measuring effect which cannot be prevented due to the laws of nature or technical limitations. Consequence: The measured values could be modified when the sample gas composition changes even when the concentration of the measured gas component remains the same.

- ▶ *If the sample gas composition has changed in such cases:* Carry out an adjustment with new test gases that correspond to the changed conditions.



This may not be necessary when the GMS800 automatically compensates such effects.

1.5

Responsibility of user**intended users**

- ▶ The GMS800 may only be installed, started and maintained by skilled technicians who, based on their technical training and knowledge as well as knowledge of the relevant regulations, can assess the tasks given and recognize the hazards involved.

Correct use

- ▶ Only use the GMS800 as described in the Operating Instructions. The manufacturer is not responsible for any other use.
- ▶ Carry out the specified maintenance work.
- ▶ Do not remove, add, or change any components in the device unless such changes are officially allowed and specified by the manufacturer. Otherwise
 - the device may become dangerous
 - the manufacturer's guarantee becomes invalid
 - the Type Examination Certificate becomes invalid (only for ATEX versions).

Special local requirements

- ▶ In addition to these Operating Instructions, observe all local laws, technical rules, and company-internal instructions valid at the site where the device is installed.

Responsibility for dangerous substances**WARNING: Mortal/health danger as a result of a gas path leakage**

When the sample gas contains noxious substances: Leaks in the gas path can be an acute danger for persons.

- ▶ Set up appropriate safety precautions (e.g. monitoring of ambient air with a gas detector, room ventilation, enclosure purging, monitoring of purge gases, instruction of personnel, emergency plan).
- ▶ Make sure the safety precautions work permanently and are complied with.

Safety precautions examples:

- ▶ Attach warning signs to the device.
- ▶ Position warning signs in the operating area.
- ▶ Inform persons who can be in the area on safety-relevant aspects.

Keeping documents

- ▶ Keep these Operating Instructions and all associated documents available for reference.
- ▶ Pass the documents on to a new owner.

1.6

Additional documents**Additional instructions and information**

These Operating Instructions include further documents specifying the technical properties of the GMS800. You need the appropriate additional documents for each device component of your GMS800.

Table 1 User documents for the GMS800 (overview)

Device component	Title	Document type
Complete device	GMS800 Series	Operating Instructions
Control unit	BCU	Supplementary Operating Instructions
	BCU – Operation with SOPAS ET	Technical Information
Enclosures	GMS810	Supplementary Operating Instructions
	GMS811	
	GMS815P	
	GMS815P-3G	
	GMS815P-PS-2G	
	GMS815P-PS-3G	
	GMS820P	
	GMS830	
	GMS831	
I/O Module	I/O Module	Supplementary Operating Instructions
Gas Module	Gas Module	
Analyzer Module	Analyzer Module DEFOR	
	Analyzer Module OXOR-E	
	Analyzer Module OXOR-P	
	Analyzer Module THERMOR	
	Analyzer Module UNOR-MULTOR	

The respective required documents are included in the scope of delivery.

Individual product information

If necessary, the GMS800 will be delivered with additional individual information:

- Product configuration (e.g. modules, system configuration)
- Recommended test gases for adjustment and factory settings
- Individual specifications where necessary.

**NOTICE:**

- Pay primary attention to the individual information and specifications delivered with the device.



If the GMS800 is delivered as part of a measuring system:

Refer to the separate documents delivered for further information.

GMS800

2 Product Description

Product identification

Functional principle

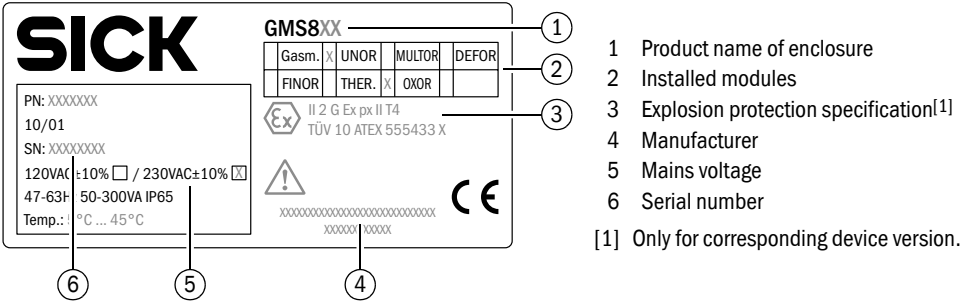
GMS800 product family

Interfaces

2.1 Product identification

Product name:	GMS800
Product variants:	→ List of additional documents (→ p. 12, § 1.6)
Manufacturer:	→ Type plate (Fig. 1)

Fig. 1 Type plate (schematic)



2.2

Functional principle/application principle

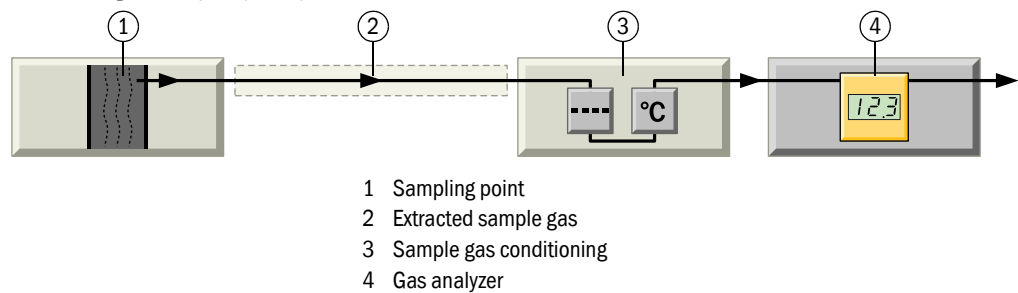
GMS800 is an extractive gas analyzer with continuous measuring operation:

- Extractive gas analysis means that a certain quantity of the gas to be analyzed is extracted from the original quantity of the gas ("sample gas" from the "sampling point") and then fed to the gas analyzer via a gas line.
- Continuous measurement means that a continuous sample gas volume flow to the gas analyzer is maintained and the gas analyzer continuously delivers measured values.
- Sample gas conditioning is required for most applications. Depending on the individual application, suitable devices can be:

Particle filters	To protect the gas analyzer's measuring system against contamination
Heated sample gas lines	To prevent condensation and ice blockages inside the sample gas path
Liquid separators	To separate any liquids or condensable components from the sample gas
Safety devices	To protect the gas analyzer and the peripheral system against each other (e.g. flame blocks in the gas path)

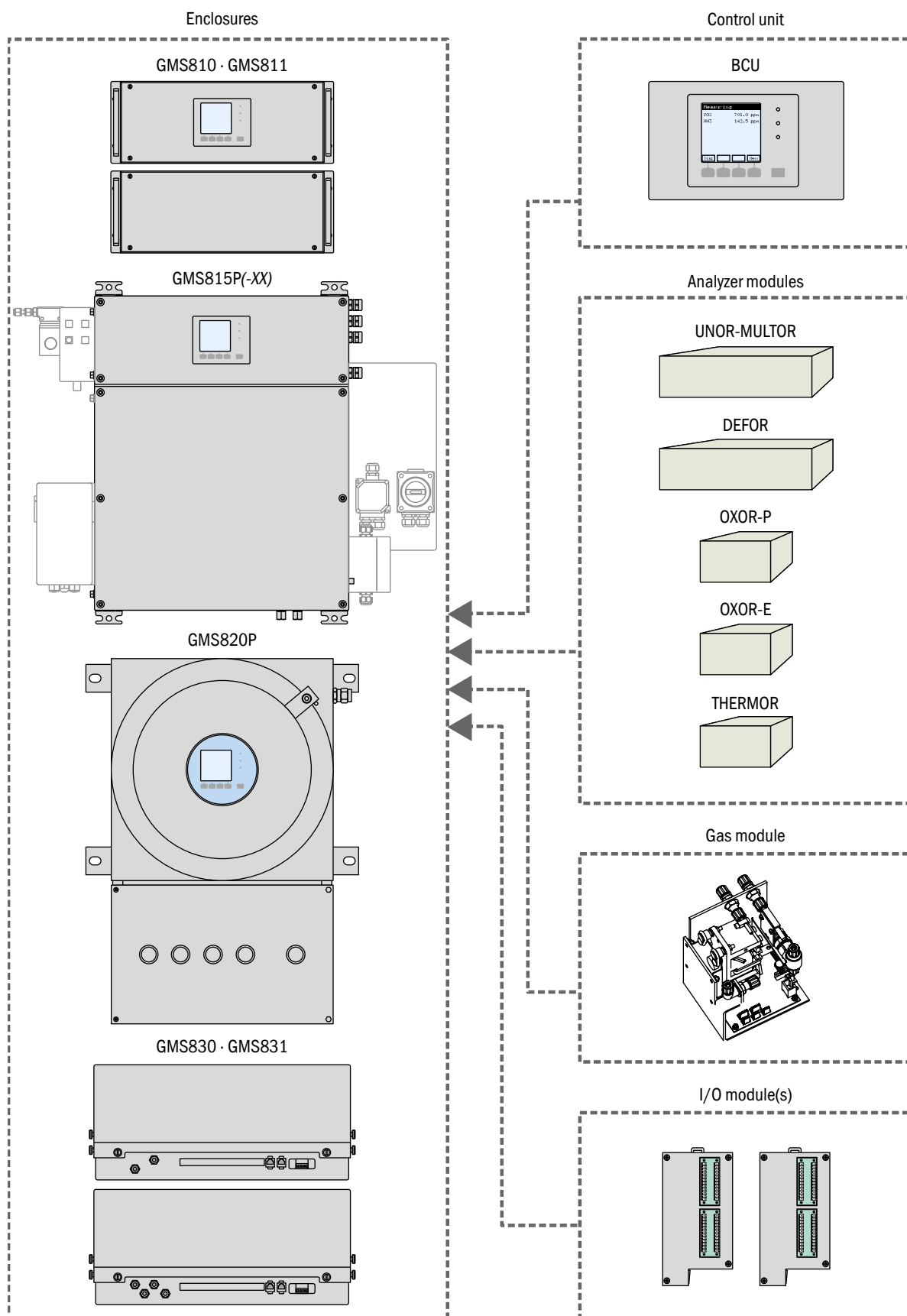
Fig. 2

Extractive gas analysis principle



Operating conditions for sample gas feed → Supplementary Operating Instructions for the Analyzer Modules fitted

Fig. 3 Product components



Subject to change without notice

2.3 Product components

2.3.1 Enclosures

Type	Intended use
GMS810	Installation in 19" rack or corresponding outer enclosure.
GMS811 [1]	Supplementation of a system with control unit. Otherwise as GMS810.
GMS815P	Wall assembly in industrial environment, standard version.
GMS815P-3G	As GMS815P, however "vapor-proof" for potentially explosive atmospheres of category "3 G".
GMS815P-PS-2G	As GMS815P, however with an enclosure pressurization system for potentially explosive atmospheres of category "2 G".
GMS815P-PS-3G	As GMS815P, however with an enclosure pressurization system for potentially explosive atmospheres of category "3 G".
GMS820P	Potentially explosive atmospheres zone 1.
GMS830	Systems with separate control unit. For large modules.
GMS831	Systems with separate control unit. For small modules.

[1] In preparation



Detailed information and specifications → Supplementary Operating Instructions for Enclosure concerned

2.3.2 Control unit

Constructive version

- Control unit integrated in enclosure.

Electronic function

- Runs as independent electronic module.
- Collects and displays measured values from other modules.
- Includes the operating and display functions.
- Controls I/O module outputs (→ p. 18, §2.3.5).
- Controls internal processes (e.g. adjustment procedure).



Detailed information → Operating Instructions for Control Unit

2.3.3 Analyzer modules

Analyzer module types

Analyzer module	Measuring principle	Measuring components, application
DEFOR	UVRAS ^[1] / UV-IFC	1 to 3 UV measuring components
OXOR-E	Electrochemical cell	O ₂ , standard requirements
OXOR-P	Paramagnetism	O ₂ , high requirements
THERMOR	Thermal conductivity	H ₂ , CO ₂ , He and others
UNOR-MULTOR	NDIR	1 to 4 IR measuring components

[1] For measuring component NO.

2.3.4 Gas module

Possible components

- Gas pump
- Pressure sensor
- Flow sensor
- Moisture sensor

Electronics

The Gas module delivers measured values and status signals of the sensors in the same manner as an Analyzer module.



Detailed information → Supplementary Operating Instructions “Gas Module”

2.3.5 I/O module

An I/O module makes the GMS800 signal connections available. The enclosure can hold 1 or 2 I/O modules (depending on device configuration).



Detailed information → Supplementary Operating Instructions “I/O Module”

2.3.6 Possible product configurations

Minimum configuration

- 1 power supply unit (provides operating voltage for internal modules)
- 1 control unit
- 1 Analyzer module with 1 measuring component

Maximum configuration

- 1 power supply unit
- 1 control unit
- 1 large Analyzer module (UNOR-MULTOR, DEFOR)
- 2 small Analyzer modules (OXOR-E, OXOR-P, THERMOR)
- 1 Gas module (gas pump, sensors)
- 1 or 2 I/O modules (per device configuration)
- Monitoring and controlling with PC + PC software “SOPAS ET”



The maximum configuration may be restricted for some enclosure types.

2.4 Information concerning the measured values

2.4.1 Physical measuring range

The “physical measuring range” corresponds to the measurement signals span created selectively by the measuring system for a gas component. These measurement signals are corrected metrologically (linearized), converted to physical units and then displayed as measured value. Further output ranges can be calculated from the physical measuring range.

The metrological specifications are valid for the respective physical measuring range. Higher measuring precision (option) can be created in the 0 ... 20% range of the physical measuring range – using additional, separate linearization of this range (option).



Specification of measuring components and measuring ranges for the individual device version → order records, delivery documents

2.4.2 Calculated measuring ranges and virtual measuring components

Several “virtual measuring components” can be set up at the factory for a single measured gas component (measuring component). Each virtual measuring component has its own measured value processing (linearization) and adjustment.

Usage

- Different measuring ranges for a measuring component are generated by setting up an own virtual measuring component for each measuring range.
- Individual measuring components can be measured with several different computations – e.g. with and without cross-sensitivity compensation. This is also performed with virtual measuring components.

Consequences

- The Measuring screens and menu functions can have several measuring components originating from the same gas component.
- Every displayed measuring component and every measuring range must be individually adjusted.



NOTICE:

To obtain a complete adjustment:

- Make a zero point and reference point adjustment individually for *each* displayed measuring component – even if the measured values originate from the same physical gas component.

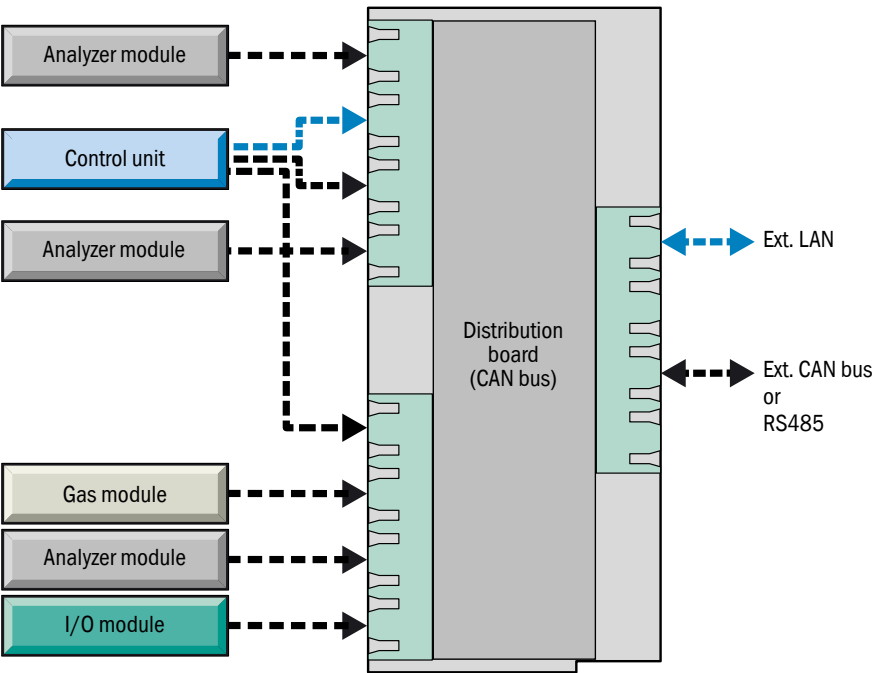
2.5 Digital interfaces

2.5.1 CAN bus

GMS800 Module data are transferred internally via a CANopen bus. Each module has an own name or module number (bus address). The control unit or PC software “SOPAS ET” communicate with each individual module.

The Analyzer modules	The control unit
● store their individual operating parameters internally (e.g. operating hours) ● automatically send their current measured values to the control unit.	● generates a status message that classifies the current measured value ● computes the measured values – as required and with appropriate programming (with other measured variables and parameters) ● displays the measured values and passes them to the outputs and interfaces.

Fig. 4 Internal connections (schematic)



Additional separator couplers must be fitted to the CAN bus when the modules are installed physically separated (e.g. in system cabinets).

2.5.2 RS485

In addition to the CANopen bus, all GMS800 modules are connected with an RS485 bus. Each GMS800 enclosure provides two RS485 connections with identical function. Several GMS800 enclosures can be interconnected to one system via the RS485 connections so that the modules of all enclosures are controlled and evaluated by one control unit. The Basic Control Unit (BCU) also uses the RS485 interface for the Modbus (→ Supplementary Operating Instructions of the Basic Control Unit (BCU)).

GMS800

3 Installation

Scope of delivery

Project planning

Connections

3.1 Scope of delivery

Item	Scope of delivery
Device	Gas analyzer, complete
	Further device components - depending on scope of order
Documentation	Operating Instructions
	Further documents - depending on device version (→ p. 12, § 1.6)
Accessories	→ Supplementary Operating Instructions for Enclosure.



NOTICE:

- ▶ Always pay primary attention to the individual information provided.



Gas connections are sealed off with plugs to protect the internal gas path from contamination, liquids and condensation.

- ▶ *Recommendation:* First remove these plugs when connecting the gas lines.

3.2 Installation/project planning guideline

Basics

Prerequisites at the installation location	→ Supplementary Operating Instructions for Enclosure.
Ambient conditions	
Gas connection details	

Required installation work

▶ Assemble/fit enclosure	→ Supplementary Operating Instructions for Enclosure.
▶ Prepare mains connection	→ p. 26, § 3.4
▶ Connect mains connection	→ Supplementary Operating Instructions for Enclosure.
▶ Make gas connections	→ p. 23, § 3.3

Additional installations as required

▶ Setting up automatic test gas feed	→ p. 25, § 3.3.6
▶ Using signal connections	→ p. 27, § 3.5



Proper installation meeting application requirements is the prerequisite for correct device functions and measuring results.

- ▶ *Recommendation:* Allow qualified skilled persons to plan and carry out installation.

3.3

Gas connections function

- ▶ Gas connections type and position → Supplementary Operating Instructions for Enclosure
- ▶ Physical conditions for the sample gas → Supplementary Operating Instructions for the Analyzer Modules fitted

3.3.1

General criteria for sample gas feed

- In most applications, peripheral components for sample gas conditioning are required in addition to the gas analyzer (e.g. dust filter, gas dryer).
- In some applications, disturbing physical effects that can falsify measured values must be considered (cross-sensitivities, absorption, adsorption, diffusion).
- The complete analysis system must be planned and installed carefully to ensure trouble-free measuring operation with low maintenance effort and excellent measuring results. This is just as decisive for measurement quality as the gas analyzer itself.



Detailed information on analysis system design → Technical Information for “Extractive Gas Analysis” (in preparation)

**WARNING: Mortal/health danger as a result of a gas path leakage**

When the unit measures toxic gases: Leaks in the gas path can be an acute danger for persons.

- ▶ Set up appropriate safety precautions (→ p. 11, § 1.5).

3.3.2

Feeding sample gas (sample gas inlet)

- ▶ Feed sample gas via the “sample gas inlet” of the enclosure.

**NOTICE:**

- ▶ Always install a fine dust filter in the sample gas feed to protect the gas analyzer against contamination.^[1]
- ▶ Prevent liquids penetrating the sample gas path in the gas analyzer.
- ▶ Prevent condensation in the sample gas path of the gas analyzer. If the sample gas contains condensable components, then only operate the gas analyzer in conjunction with a sample gas conditioning system
→ Technical Information for “Extractive Gas Analysis”^[2]
- ▶ *Before feeding sample gas:* Check whether the sample gas can chemically attack the sample gas path materials.
→ Supplementary Operating Instructions for the Analyzer Modules fitted

[1] *Even when the sample gas is free from particles:* Fit a dust filter as safety filter so that the gas analyzer is protected during operational malfunctions or defects.

[2] In preparation.

**WARNING: Hazards from noxious sample gases**

- ▶ *When the sample gas contains substances dangerous to health:* Check whether additional safety precautions are necessary (→ p. 11, § 1.5).

3.3.3

Channeling off exhaust gas (sample gas outlet)

- Connect the “sample gas outlet” to a suitable collection point (e. g. exhaust duct).

**CAUTION: Risk of health risks/damage**

If the exhaust gas can create condensate: It is possible that acid is created in the exhaust gas line. Acid could be noxious and corrosive.

- Safely collect and dispose of condensate.
- Prevent condensate penetrating the gas analyzer.

**CAUTION: Risk of erroneous measurements**

The sample gas must not penetrate the enclosure.

- Channel the sample gas off safely from the sample gas outlet. Otherwise significant measurement errors could occur.

If the GMS800 is not fitted with the “sample gas pressure compensation” option:

**CAUTION: Risk of erroneous measurements**

- Sample gas outlet must not be throttled.
- No significant counterpressure at the sample gas outlet.
- No strong pressure fluctuations may occur at the sample gas outlet.
- Make sure the sample gas can flow out “freely”.
- Install regulating valves to adjust the volume flow only before the sample gas inlet.

3.3.4

Feeding span gas (option)

Only applicable for device versions with “flowing span gas”

Devices with span gas connections have a second internal gas path through which the span gas should flow.

- Feed span gas via the span gas inlet. Maintain the same operating conditions as on the sample gas inlet.
- Channel the span gas outlet off to a suitable collection point. Maintain the same operating conditions as on the sample gas outlet.



The span gas must be used as “zero gas” during adjustment so it can be advantageous to install a connecting line from the span gas inlet to the sample gas inlet.

3.3.5

Providing special gas connections (only for special versions)

Special versions of the GMS800 may possibly be fitted with individual gas connections (e.g. for a second sample gas path).

- Observe the individual information provided.

3.3.6

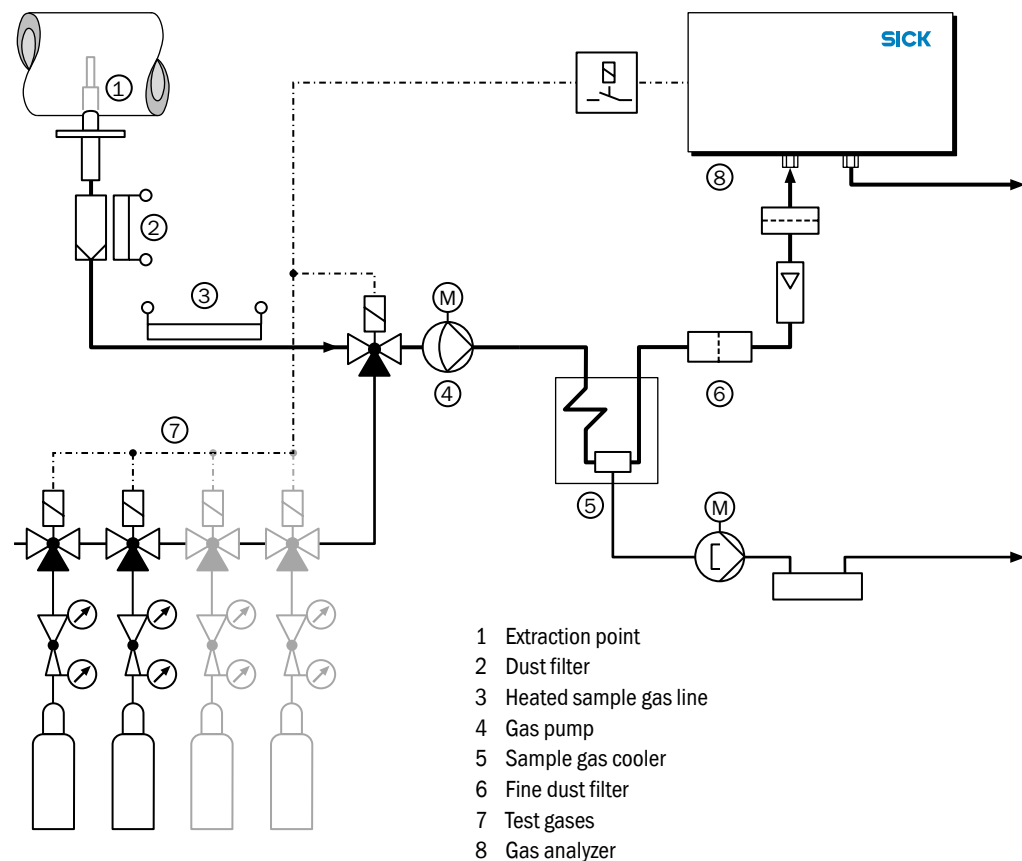
Providing installations for test gases (when required)

When cyclic adjustments are to be setup or when the test gases are to be fed automatically during adjustments:

- ▶ Make the required test gases (→ p. 41, §6.3) available in pressure cylinders or from pressure lines.
- ▶ Install suitable pressure reducers to provide the correct feed pressure (→ Supplementary Operating Instructions for the Analyzer Modules fitted).
- ▶ Install solenoid valves or similar devices for electric control of the test gas feed.
- ▶ Configure the GMS800 digital outputs to control the solenoid valves (→ Supplementary Operating Instructions for "I/O Module").
- ▶ Connect the solenoid valves to the digital outputs.
- ▶ Assign the test gases to the suitable digital output in the Test Gas Table (→ Technical Information for Control Unit).

Fig. 5

Installations for gas feed (example for emission measurement)



3.4 Mains connection

3.4.1 Safety information for mains connection

- ▶ Check whether the available mains voltage matches the type plate specifications. *If this is not the case:* Do not connect the device.
- ▶ Check whether the mains supply has a functional protective conductor (PE). *If this is not the case:* Do not connect the device.



CAUTION: Health risk

- ▶ Only connect the device to a mains supply with a functional protective conductor (protective earth, PE).
 - ▶ Only start the device when a correct protective conductor is installed.
 - ▶ Never interrupt a protective conductor connection (yellow-green cable) inside or outside the enclosure.
- Otherwise electric safety is not ensured.

3.4.2 Install an external mains fuse

- ▶ Install a mains fuse in the mains supply. Recommended fuse value for an external device: T 16 A.



The GMS800 needs more power (“inrush current”) than the rated current temporarily when switched on. Reference value: 30 A at 230 VAC mains voltage (60 A at 115 VAC).

- ▶ Use fuses with slow trigger characteristic.



Internal mains fuses:

- *Primary:* Fuse in internal power supply unit (6.3 A). – *If this fuse triggers:* Replace the complete power supply unit.
- *Secondary:* Fusible cutout on the internal “distribution board” (on CANopen connections). – *If this fuse triggers:* Clear the malfunction cause and renew the fusible cutout.^[1]

[1] Fuse element F10A 250V D5x20, Part No. 6044838.

3.4.3 Install an external mains switch

- ▶ Install a mains disconnect switch close to the device to be able to switch the mains supply to the device on and off.
- ▶ Mark this mains switch clearly and unmistakably.



European standard EN 61010 specifies that any fix-mounted devices not having an own mains switch must be equipped with an external mains switch.

3.4.4 Connect mains connection

→ Supplementary Operating Instructions for Enclosure.

3.5 Signal connections

3.5.1 Safety information for signal connections

Safe input signals


NOTICE:

Signals connected to signal connections shall be low voltage (max. 30 Vac or 60 Vdc), derived from: a secondary circuit, double or reinforced insulated from mains, e.g. a "SELV" circuit per IEC 60950-1.

Install disconnected from the mains


NOTICE: Sensitive electronics

Before signal connections are established (also with plug connections):

- ▶ Disconnect the GMS800 and all connected devices from the mains (switch off).

Otherwise the internal electronics could be damaged.

Adjustment procedure safeguard

While an adjustment procedure is running, status "Functional control" is activated and the test gas measured values output as measured values.


CAUTION: Risk during adjustments

Test gas values are output as measured values during an adjustment procedure.

- ▶ Check whether the "Function control" digital output must be processed or displayed at external stations.
- ▶ Install the "Function control" digital output accordingly when necessary.

Otherwise the measured values output from the test gases could possibly create dangerous or undesired situations.

3.5.2 Information in other documents (notice)

Signal connections ...	See for information
On an internal I/O module	→ Supplementary Operating Instructions for "I/O Module"
On other external components	→ Relevant separate information

Interfaces



Position of interface connections → Supplementary Operating Instructions for Enclosure

**NOTICE:**

Signals connected to interfaces shall be low voltage (max. 30 Vac or 60 Vdc), derived from: a secondary circuit, double or reinforced insulated from mains, e.g. a "SELV" circuit per IEC 60950-1.

Ethernet

A PC can be connected to the Ethernet interface (network connection). The "SOPAS ET" application program supports digital communication with the GMS800.

Application options with "SOPAS ET":

- Measured value and status inquiries
- Remote control
- Parameter setting
- Diagnosis
- Setting the internal configuration

CAN bus

External System modules can be connected to the CANopen interfaces. One of the CANopen connections is reserved for the CAN bus terminator (terminating resistor).

RS485

Several GMS800 enclosures can be interconnected to one system via the RS485 connections.

- *When the delivered GMS800 configuration has several enclosures:* Observe the individual information provided.



The Basic Control Unit (BCU) uses the RS485 interface also for the Modbus (→ Supplementary Operating Instructions of the Basic Control Unit (BCU)).

GMS800

4 Start-up

Safe preparation
Switch-on procedure
Individual adaptations

4.1

Safety information on start-up**NOTICE: Risk of damage**

Liquids and particles (dust) must not penetrate the gas analyzer measuring system. The gas analyzer is normally unusable when liquids or particles have penetrated the measuring system.

Before activating sample gas feed to the gas analyzer:

- ▶ Make sure no liquids (e.g. condensate) or particles can penetrate the gas analyzer.
- ▶ Check whether sample gas feed to the gas analyzer functions correctly (e.g. dust filter, valves).

Possible single measures:

- ▶ Wait until system components that remove condensable substances out of the sample gas ^[1] are in the operating state (e.g. sample gas cooler).
- ▶ Wait until heated system components ^[1] have reached operating temperature (e.g. heated sample gas line).

[1] If fitted.

4.2

Start-up procedure**Before GMS800 start-up**

- 1 Wall enclosures and Ex-d enclosures ^[1] in potentially explosive atmospheres: Close the enclosure and check the enclosure for leak tightness.
- 2 Check gas feed condition and leak tightness.

When corresponding devices are fitted:

- 1 Put devices for sample gas conditioning into operation (e.g. gas cooler) and/or check their condition (e.g. filter).
- 2 Check pressure reserves in test gas cylinders.
- 3 Put protection devices into operation (e.g. enclosure purging).
- 4 Wait until all devices are ready for operation.

Start up the GMS800

- Switch mains supply on (→ Supplementary Operating Instructions for Enclosure).

Wait until ready for operation

- 1 Wait until the control unit is ready for operation (→ Operating Instructions for Control Unit).
- 2 Wait until the GMS800 is ready for operation. This is the case when no malfunctions are displayed after the heating up phase.
- 3 Start sample gas feed (e.g. open valve).



- Heating up time: $\approx 0,7 \dots 2$ hours (depending on ambient temperature)
- Each module has a function in the Menu system which shows the module state with LED symbols.

4.3

Measures after start-up

- Carry out an adjustment (→ p. 37, §6).

[1] In preparation

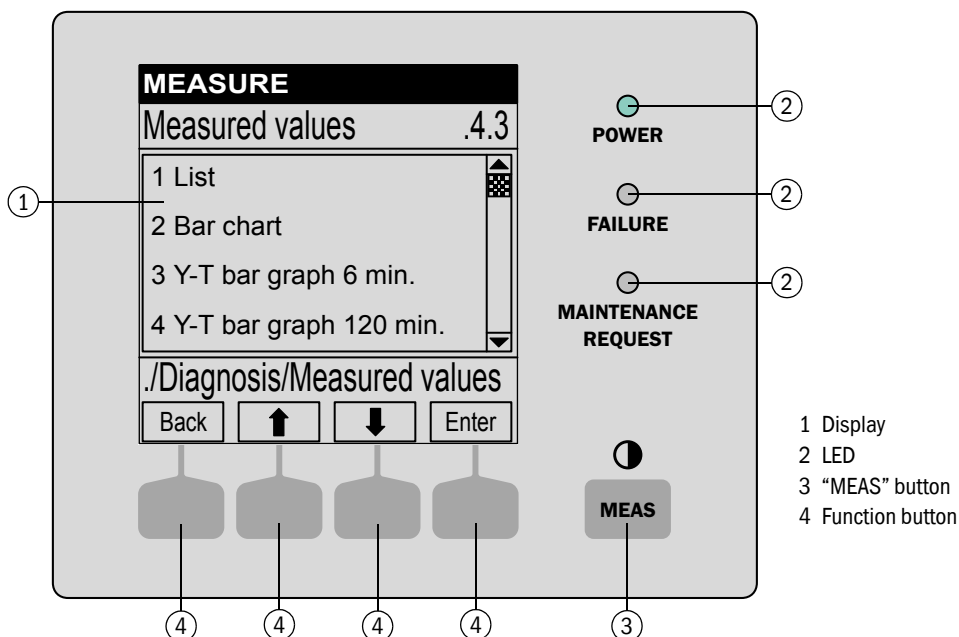
GMS800

5 Operation

- Operating elements
 - Menu system
- Operating state recognition
- Behavior in an emergency

5.1 Operating and display elements (quick guide)

Fig. 6 Basic Control Unit (BCU) operating and display elements



- Function of LEDs → Supplementary Operating Instructions "BCU"
- The "MEAS" button starts the measured value display immediately.
- The respective function of the function buttons is shown on the display.



Detailed instructions → Supplementary Operating Instructions "BCU"



The display lighting can possibly switch off automatically after a certain time (→ Supplementary Operating Instructions "BCU").
 ► To reactivate the display lighting: Touch the left or right function button.

5.2 Menu system

5.2.1 Variants of the menu system

The menu functions of the Analyzer modules and of the Gas module are available in two variants:

- Menu system in the Basic Control Unit (BCU)
- Menu system in the PC software "SOPAS ET"

The PC software "SOPAS ET" has more complex menu functions than the Basic Control Unit (BCU).

5.2.2 User levels

Certain menu functions are only available when user level "Authorized operator" is active.



Trained and authorized skilled persons can make advanced parameter settings at "Service" level.

5.3 Checking the operating state (visual check)

5.3.1 Recognizing a safe operating state

Control unit of gas analyzer

- Operation display of control unit is on
- No malfunction messages on display
- Measured value display with normal background color
- Measured values in normal (expected) range

Gas analyzer peripherals

- Gas feed functioning correctly (e.g. pump, filter)
- Peripheral devices functioning correctly (e.g. fan, heater)

5.3.2 Recognizing an unsafe operating state

Complete device

- Unusual smell (gas, smoke, heat)
- Heavy damage or deformation of enclosure
- Defective or damaged connections or connection lines
- Unusual noises



Some Analyzer modules create rhythmic operating noises.

Control unit

- Operation display not on
- Malfunction message on display



- A malfunction message appears on the display during the heating up phase after switching on. The GMS800 is not ready for operation in this state. This is however not an unsafe operating state.
- An "Alarm" message is not an indication for an unsafe operating state.



An "Alarm" message signals that the measured value has exceeded a programmed limit value.

- *When the GMS800 reports an "Alarm":* Check whether the current measured value requires an operational reaction.

Peripherals

- Leak in a gas line
- Incorrect operating conditions (e.g. ambient temperature, gas pressure)
- Heat build-up (ambient temperature too high)
- Moisture condensation/moisture on enclosure
- Peripheral device down (e.g. fan, heating)



CAUTION: Danger caused by unsafe operating state

When the GMS800 is or could be in an unsafe state:

- Put the GMS800 out of operation, disconnect from the mains voltage and signal voltage and secure against unallowed or accidental start-up.

**WARNING: Hazards through a gas leak**

- *When gas escapes uncontrolled:* Immediately check whether the gas can be dangerous to health or combustible.
If this is the case: Immediately follow the local Operating Instructions that govern behavior during uncontrolled escape of gas.

5.4

Behavior in an emergency**Fire:**

- 1 Stop gas feed to the GMS800.
- 2 Disconnect the GMS800 from the mains voltage (mains switch or emergency switch).
- 3 Switch off possible peripheral devices (e.g. heater).
- 4 Trigger an alarm/make an emergency call.
- 5 Follow the local Operating Instructions governing behavior when fire breaks out.
- 6 Inform the fire brigade about dangerous gases when necessary.

Unsafe operating state (→ p. 35, § 5.3.2):

- 1 Stop gas feed to device.
- 2 Disconnect the device from the mains voltage (mains switch or emergency switch).
- 3 Prevent unauthorized or unintentional start-up.
- 4 Protect the measuring system against condensation and penetration by liquids.

If a protection device is defective (when fitted):

- 1 Disconnect the GMS800 from the mains voltage (mains switch or emergency switch).
- 2 Stop gas feed to the GMS800.
- 3 Prevent unauthorized or unintentional start-up.
- 4 Protect the measuring system against condensation and penetration by liquids.

GMS800

6 Adjustment

Introduction

Guideline

Preparations

Adaptations

Procedures

6.1 Adjustment introduction

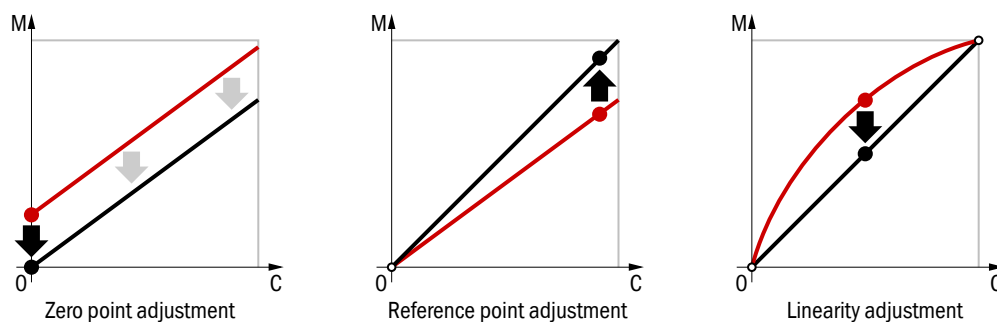
6.1.1 Adjustment purpose

It is unavoidable that some of the properties of Analyzer modules will change during the operating time. This changes measurement results even when external conditions remain identical. This gradual change in measured results is known as drift. Zero point drift and reference point drift exist. These drifts are measured during adjustment and the relation between real concentration and measured value (characteristic curve) is corrected accordingly (→ Fig. 7).

The linearity of the characteristic curve (proportional relation between real value and measured value) can also be corrected afterwards .

Fig. 7

Adjustment functions (schematic)



6.1.2 Principle adjustment procedure

- 1 A test gas is fed in.
- 2 A measured value (actual value) is determined with this test gas.
- 3 This actual value is then compared against the programmed nominal value.
- 4 The internal adjustment parameters are then corrected mathematically so that the actual value then corresponds to the nominal value.

To attain complete adjustment, this procedure must be performed twice for each measuring component – once for the zero point and once for the reference point. Sequence control programs control these procedures (→ Technical Information for Control Unit).

6.1.3

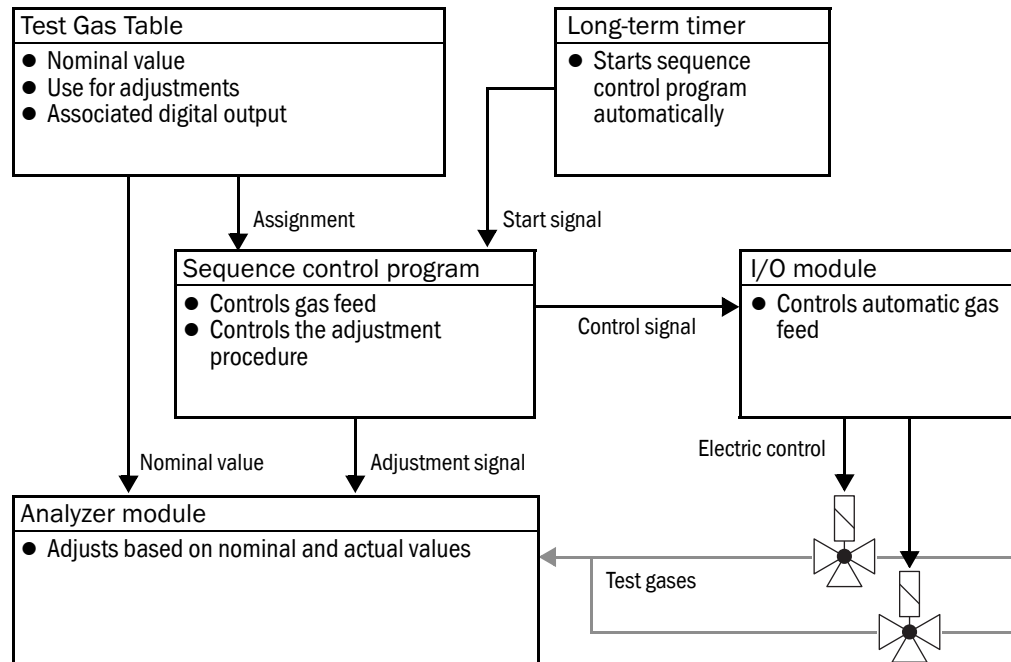
Internal organization of adjustment procedures

Three internal instances are decisive for adjustment:

- The Test Gas Table – to program test gas settings
- Sequence control programs for adjustment
- Long-term timer – for time-controlled automatic starts of sequence control programs

Fig. 8

Internal organization of adjustment procedures



6.2 Adjustment guideline

6.2.1 How often do you have to adjust?

The GMS800 should be adjusted

- after start-up
- at regular intervals during operation (weekly to monthly).



- Pay primary attention to the adjustment information in the Supplementary Operating Instructions for the Analyzer Modules fitted.



- Choose larger adjustment intervals when possible (e.g. 3 or 6 months) when permitted by the application or when expressly allowed (e.g. in a TÜV approval).
- Specialized measuring systems (e.g. process applications with complex gas conditioning systems) may need a different adjustment concept.

6.2.2 What do I need for an adjustment?

To perform an adjustment, you need:

- For each measuring component of the GMS800
 - suitable zero gas (→ p. 41, §6.3.1)
 - suitable reference gas (→ p. 42, §6.3.2)
- Time in which normal measuring operation may temporarily stop.

Further prerequisites are:

- Correct test gas parameter settings^[1]
- Correct runtime settings ^[1]

The GMS800 can control test gas feed automatically.^[1]

6.2.3 How can you perform an adjustment?

You can use the following alternative adjustment procedures:

	Alternative adjustment procedures	Requirements	See
A	Single comparisons with manual test gas feed	Suitable test gas settings	→ Operating Instructions for Control Unit
B	Single comparisons with automatic test gas feed	As in [A] + installations with automatic test gas feed	→ p. 25, §3.3.6
C	Automatic adjustment started manually	As in [B] + selection of suitable sequence control program	→ Technical Information for Control Unit
D	Fully automatic (cyclic) adjustment	As in [C] + programmed cyclic trigger	

[1] → Technical Information for Control Unit

6.3 Test gases

**NOTICE:**

- Pay primary attention to the information and specifications on test gases in the Supplementary Operating Instructions for the Analyzer Modules fitted.

6.3.1 Zero gas

General requirements

Zero gas should not normally cause a measuring effect for the measuring components for which the metrological zero point is adjusted with this gas (nominal value: "0"). A zero gas cannot therefore contain the measuring components.



- The same zero gas can be used for all measuring components in most applications.
- Nitrogen (N₂) is normally used as zero gas with quality "technical" or "top grade" depending on the application.
- Filtered atmospheric fresh air can be used as zero gas for some applications.

Application-specific zero gas

A definite nominal value can also be set for zero gas. This means a zero gas that causes certain measuring effects can be used in special applications. These effects must be known quantitatively and considered appropriately when setting the zero gas nominal value.



- Pay primary attention to the individual information provided on zero gas.

6.3.2

Span gases**General requirements**

The reference point or the linearity are adjusted with span gases. A span gas is basically a mixture of zero gas and a measuring component whose measurement is to be adjusted.

Span gas mixtures

In many applications, you can also use span gas mixtures with more than one measuring component to adjust the reference point of several measuring components.

However, please note that span gas mixtures should *not* be used in the following applications:

- When the coexistence of the gas components could physically interfere with the gas analysis
- When the gas components could chemically react with each other
- When the mixture components in the GMS800 create cross-sensitivity effects in those measuring components to be adjusted and these cross-sensitivity effects cannot be automatically compensated
- When separate information was delivered with the analyzer which rules out the use of span gas mixtures.

Appropriate nominal values

The nominal value of a span gas is the actual concentration of the measuring component in the span gas.

- *For reference point adjustment:* On the GMS800, the nominal value may amount to 10 ... 120% of the end value of the respective physical measuring range. Adjust the nominal value in the range 65 ... 100% of the physical measuring range for precise adjustment.
- *For linearity adjustment:* The nominal value should be approx. 50% (40 ... 60%) of the end value of the respective physical measuring range.



- ▶ Pay primary attention to the span gas information in the Supplementary Operating Instructions for the Analyzer Modules fitted.
- ▶ Pay primary attention to the individual information provided.

6.3.3

Physical conditions for test gases**Principle**

Test gases should be fed into the gas analyzer under the same conditions as the sample gas.

- ▶ *When devices are available for sample gas conditioning (e.g. filter):* Let the test gases flow through the sample gas conditioning before reaching the gas analyzer.
- ▶ *If a sample gas cooler is used:* Observe §6.3.4 (→ p. 44).

Volume flow

- ▶ Set the volume flow (flow) of the test gases so that it more or less corresponds to the sample gas volume flow.

Feed pressure

- ▶ *Without a sample gas pump:* Feed the test gases with the same primary pressure as the sample gas.
- ▶ *With a sample gas pump (option in Gas module):* Feed the test gases with low overpressure (+50 ... +100 mbar). Set the overpressure so that the volume flow is as large as the sample gas volume flow during operation.

**NOTICE:**

On devices with a sample gas pump:

- ▶ Make sure the feed pressure of the test gases is limited (check pressure regulator).

Otherwise the fitted sample gas pump can be damaged.

6.3.4

Test gas feed with sample gas cooler

Only applicable for applications where a sample gas cooler is used.

Adjustment with “dry” test gases

In the method with “dry” test gases, the test gases flow from the source (pressure cylinder) directly into the gas analyzer without flowing through the sample gas cooler.



Possible methods to determine the sample gas cooler influence:

- 1 Feed a test gas directly into the gas analyzer (as during adjustment). Note the measured value displayed for this test gas.
- 2 Feed the same test gas again through the sample gas cooler (as with the sample gas) before it reaches the gas analyzer. Note the measured value.
- 3 Consider the difference between the two measured values during measuring operation.
- 4 Possibly repeat this comparison measurement regularly.

Advantages:	● The physical conditions are identical during adjustment. This means the adjustment results can be compared directly. ● This method allows following the gas analyzer drift.
Disadvantages:	● The influence of the sample gas cooler is not considered during adjustment. ● It may be necessary to quantify the influence of the sample gas cooler.

Adjustment with “wet” test gases

The test gases are subjected to the same influences as the sample gas when the test gases flow through the sample gas cooler before reaching the gas analyzer. The result is “wet” test gases with the same H₂O content as the sample gas.

Advantages:	● The actual influence of the sample gas cooler is recorded physically and considered in the adjustment.
Disadvantages:	● Because the physical conditions in the sample gas cooler are not exactly constant, the results of single adjustments are also not exactly identical. This must be considered when assessing the drift. ● Due to the fact that adjustment gases from gas cylinders practically do not contain any H₂O, the sample gas cooler can dry out during the course of a long adjustment procedure. This would neutralize the advantages of this method.

GMS800

7 Shutdown

Protective measures
Switch-off procedure
Disposal

7.1 Safety information on shutting down



NOTICE:

The Analyzer modules are heated to maintain constant internal temperatures. This also prevents condensation occurring in the measuring system during operation. However, when the gas analyzer is shutdown condensation could occur in the Analyzer modules as they cool down. This can damage the Analyzer modules or make them unusable. – Therefore:

- ▶ Always purge the internal sample gas path thoroughly with a “dry” neutral gas before each shutdown task.



WARNING: Health hazards through dangerous gases

If the GMS800 is used to measure toxic or dangerous gases:

- ▶ Thoroughly purge all sample gas paths with a neutral gas (e.g. nitrogen) before opening sample gas paths or components carrying sample gas.

7.2 Preparations for shutdown

7.2.1 Secure connected stations



- Gas analyzer shutdown can affect external stations. It may be necessary to take the switching logic used by the switching outputs of the GMS800 into consideration (→ Supplementary Operating Instructions for “I/O Module”).
- If a data processing system is connected, it may be required to manually indicate a planned shutdown, so that the system will not interpret the shutdown as a gas analyzer malfunction.

- ▶ If required, inform connected external stations that a shutdown is planned.
- ▶ Ensure operational safety is not endangered by shutting down, e.g. when the gas analyzer is used to monitor processes or atmospheres.
- ▶ Ensure shutdown does not unintentionally trigger an automatic emergency signal.

7.2.2 Purge sample gas out of the gas analyzer

- 1 Stop gas feed to the GMS800.
- 2 Disconnect the GMS800 from external sample gas paths so that no sample gas can flow into the GMS800.
- 3 Purge all gas paths in the GMS800 for several minutes with a “dry” neutral gas – for example with technical grade nitrogen or with a zero gas. Possibly include peripheral gas paths in this purging operation.
- 4 Then close off all GMS800 gas connections or close the related valves in the purged gas path.

7.2.3 Deactivate enclosure pressurization (when fitted)

When the enclosure is fitted with an active system for enclosure pressurization (e.g. inert gas purging):

- ▶ Shut the enclosure pressurization system down (→ Operating Instructions for the system involved).

7.3

Switch-off procedure

- 1 Carry out shutdown preparations (→ p. 46, § 7.2).
- 2 Disconnect the GMS800 mains supply at an external source (external mains switch).

**WARNING: Explosion hazard in potentially explosive atmospheres**

If the device is installed in a potentially explosive atmosphere:

- ▶ *Before opening the enclosure:* Wait for the prescribed delay times to elapse (→ Supplementary Operating Instructions for Enclosure or Operating Instructions for Enclosure Pressurization System).

7.4

Protective measures for long-term storage

- ▶ *When the GMS800 has been separated from gas lines:* Close off the GMS800 gas connections (with sealing plugs, if necessary with adhesive tape) to protect against moisture, dust or dirt penetrating the internal gas path.

When the GMS800 is fitted with the OXOR-E Analyzer module: Keep gas connections gas-tight during storage.



The service life of the OXOR-E Analyzer module will be reduced by contact to oxygen in the air even when the device is switched off.

- ▶ Cover open electrical connections dust-tight, e.g. with adhesive tape.
- ▶ Protect the keypad and display against sharp-edged objects. If necessary, cover the instrument with protective material (e.g.: Cardboard, styrofoam).
- ▶ Select a dry and well-ventilated room for storage whenever possible.
- ▶ Pack the device (e.g. in a plastic sack).
- ▶ *When high air humidity can be expected:* Include a drying agent (SilicaGel) in the packing.

7.5

Transport**CAUTION: Risk of injuries and accidents**

- ▶ Observe the safety information for transport (→ Supplementary Operating Instructions for Enclosure).

- ▶ Protect the enclosure before transport (→ § 7.4).
- ▶ Use the original packaging for transport whenever possible.
- ▶ A transport container with adequate stability can also be used. Protect the device with padding from shocks and vibrations and thoroughly secure the device in place inside the transport container. Make sure there is sufficient clearance from the walls of the container.



Accompanying documents when shipping for repairs → § 7.6

7.6

Shipping for repair

When the device is being sent to the manufacturer or a Service company for repairs:

Please enclose the following information so that the device can be repaired as quickly as possible:

- ▶ An error description, as precise as possible (meaningful keywords suffice).
- ▶ *For unclear functional problems:* A short description of the operating conditions and installations (connected devices etc.).
- ▶ *If shipping was agreed with the manufacturer:* The contact person at the manufacturer's who is informed about the matter.
- ▶ A contact person in the user's company (for possible questions).



Please add the information even if your matter has already been discussed with a manufacturer's employee.

7.7

Disposal**CAUTION: Risk of environmental hazards**

- ▶ Observe the information in these Operating Instructions.
- ▶ Observe local regulations and laws on disposal of industrial waste and electric devices.

**WARNING: Health hazards through dangerous gases**

If the GMS800 is used to measure toxic or dangerous gases:

- ▶ Thoroughly purge all sample gas paths with a neutral gas (e.g. nitrogen) before opening sample gas paths or components carrying sample gas.

These subassemblies could contain materials which may require special disposal:

- *Sample gas paths:* Toxic materials in the sample gas could have been absorbed or trapped in the »soft« gas path material (e.g. hoses, sealing rings).
- *Sample gas filter:* Sample gas filters can be contaminated with pollutants.
- *Electronics:* Electrolyte capacitors, tantalum capacitors
- *Display:* Liquid of the Liquid Crystal Display (LCD)

GMS800

8 Maintenance

Maintenance plan

Visual inspection

Cleaning

Leak tightness check

8.1 Maintenance plan

8.1.1 Maintenance by the user

Maintenance interval ^[1]				Maintenance work	Information	Note
1D	1W	1M	6M			
(□)	□	□	□	► Make a visual inspection	→ p. 51, § 8.2	
	(□)	□	□	► Perform adjustment	[2]	a
	(□)	(□)	(□)	► Check/clean gas supply and discharge lines [3]		a b
			□	► Check operating hours of DEFOR Analyzer module [4]		

[1] D = day(s), W = week(s), M = month(s).

[2] → Operating Instructions for Control Unit.

[3] If fitted.

[4] The UV lamp in the DEFOR Analyzer module must be replaced about every 2 years (→ § 8.1.2). With measuring component NO: The gas filter fitted for NO measurement must be replaced every 2 years approximately.

Note	Explanation
a	Maintenance interval depends on the individual application
b	Only when solids deposit in the gas lines – as required



► In addition, observe all official and internal company regulations valid for the individual application.

8.1.2 Maintenance by Service technician

Maintenance interval ^[1]				Maintenance work	Note
6M	1Y	2Y	10Y		
□	□	□	□	► Check/service fitted gas pump [2]	a
□	□	□	□	► Check function of flow sensor [3]	a
□	□	□	□	► Check important operating functions (e.g. alarm message)	a
(□)	□	□	□	► Readjust adjustment unit [4]	
	□	□	□	► Adjust H ₂ O measurement [5]	
	□	□	□	► Replace OXOR-E module [6]	a
	□	□	□	► Check gas paths for leak tightness	
		□	□	► Replace UV lamp [7]	
		□	□	► Replace gas filter for NO measurement [8]	
			□	► Replace battery in control unit	c

[1] M = month(s), Y = year(s).

[2] Only for devices with Gas module with gas pump.

[3] Only for devices with Gas module with flow sensor.

[4] Only for devices with DEFOR Analyzer module with adjustment unit (option).

[5] Only for devices with H₂O measurement.

[6] Only for devices with OXOR-E Analyzer module.

[7] Only for devices with DEFOR Analyzer module

[8] Only for devices with DEFOR Analyzer module and measuring component NO.

Note	Explanation
a	Maintenance interval depends on the individual application
c	Carry out at the manufacturer's factory or workshop with suitable equipment

8.2

Visual inspection**Maintenance interval**

Recommendation: Max. 2 days

Procedure

- 1 Check the GMS800: Inspect the control unit display (no malfunction fields).
- 2 Check test gas supply (when present):
 - Remaining reserves in pressure cylinders
 - Feed pressure
 - State of gas lines and valves
- 3 Check peripheral installations (when present), e.g.:
 - Gas sampling probe
 - Sample gas line (state, connections)
 - Particle filter (dust filter)
 - Protective filter (e.g. corrosion inhibitor filter)

8.3

Cleaning of the enclosure

- ▶ To clean the enclosure, use a soft cloth.
- ▶ If required, wet the cloth with water and a mild cleaning solution.
- ▶ Do not use any mechanically or chemically aggressive cleaning agents.
- ▶ Do not allow any liquids to penetrate the enclosure.

**CAUTION: Hazardous situation if liquids penetrate the enclosure**

If liquids have penetrated the device:

- ▶ Do not touch the device any more.
- ▶ Shut the device down immediately by disconnecting the mains voltage at an *external station* (e.g. pull out the mains cable at the socket or switch off the external mains fuse).
- ▶ Contact the manufacturer's customer service or other trained skilled persons able to repair the device.

8.4 Leak tightness check of sample gas path

8.4.1 Safety information on gas leak tightness



WARNING: Hazards through leaky gas path

- There is a risk to health when the gas path is leaky when using toxic sample gases or sample gases hazardous to health.
- If the sample gas is corrosive or can create corrosive liquids with water (e.g. air moisture), there is a risk of damage to the gas analyzer and neighboring equipment when the sample gas path is leaky.
- If the escaping gas is explosive or can create an explosive gas mixture together with ambient air, there is a *risk of explosions* when the safety measures on explosion protection are not implemented.
- Measured values can be erroneous when the gas path is leaky.



If the gas path is determined to be leaky:

- ▶ Shut off gas feed.
- ▶ Put the gas analyzer out of operation.
- ▶ *If the escaping gas can be dangerous to health, corrosive or explosive:* Remove the escaping gas systematically (purge, suction off, vent) whilst maintaining the necessary safety measures, e.g. for
 - explosion protection (e.g. purge enclosure with inert gas)
 - health protection (e.g. use breathing protection)
 - environment protection.

8.4.2 Test criterion for gas leak tightness

- At the most, 10 ml/min (0.6 l/h) may escape from the internal gas path of the gas analyzers (gas inlet to gas outlet) with an overpressure of 100 kPa (1 bar) against the ambient air pressure. The device is classified as leaky when the leakage rate is higher.
- Recommended test interval: Max. 6 months

8.4.3 Simple test method for gas leak tightness



Other test methods are also available - e.g. using a controllable mass flow rate controller.

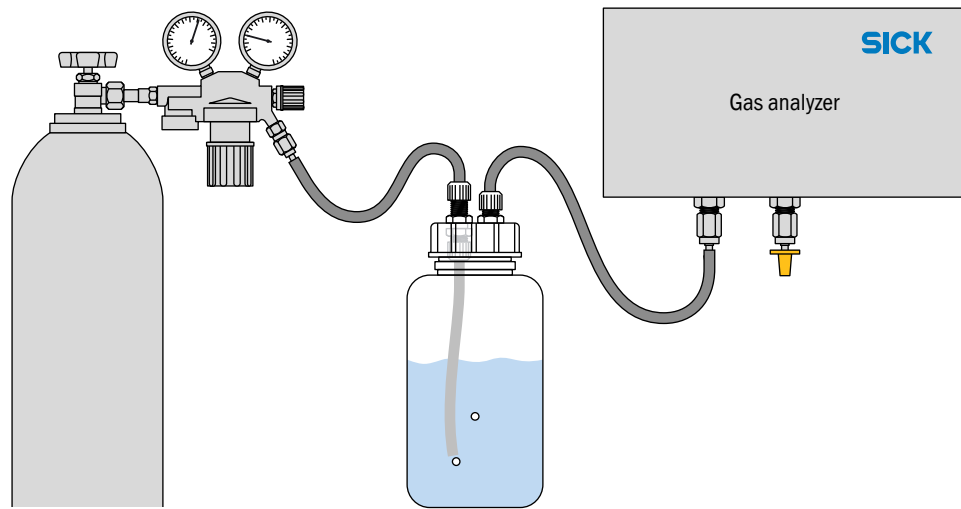
Test equipment

For a simple test, you need

- A compressed gas cylinder with adjustable pressure reducer (recommendation: Nitrogen)
- A »washing bottle« with two hose connections (→ p. 53, Fig. 9).
 - The washing bottle must be able to withstand the test pressure (1 bar) and be closable gas-tight.
 - The hose leading (or a suitable tube) into the water should have an inner diameter of 5 mm (outlet opening diameter).
 - Plain water can be used as filling. Only use enough water so that no water can escape via the gas outlet of the washing bottle.

Fig. 9

Simple test method for leak tightness check (example)



Test procedure



When the gas analyzer has several separate internal gas paths:

► Perform this procedure individually for each gas path.

- 1 Put the gas analyzer out of operation. Separate the gas inlet and outlet of the gas analyzer from the installation (when fitted).
- 2 Connect the gas analyzer gas inlet to the washing bottle gas outlet.
- 3 Close the gas analyzer gas outlet off gas-tight, e.g. with a sealing plug.
- 4 Also close off all the other internal gas path connections (as available) in the same manner.
- 5 Check: The valve on the pressure reducer gas outlet must be closed off. Now open the main valve of the compressed gas cylinder.
- 6 Adjust the pressure reducer so that the output pressure (secondary pressure) is 150 kPa (1.5 bar).
- 7 Connect the pressure reducer gas outlet to the washing bottle gas inlet.
- 8 Open the pressure reducer valve *slowly* (avoid sudden pressure increase).
- 9 Wait until the pressure ratio is constant (several seconds).
- 10 Observe the washing bottle: Count the number of air bubbles rising for 1 minute.
The gas path is classified as tight when less than 60 air bubbles rise per minute.
- 11 To terminate the test procedure:
 - Close off the valve on the pressure reducer gas outlet.
 - To release the gas pressure: Carefully loosen the connection hose *on the washing bottle gas outlet*.
 - Reconnect the gas connection on the gas analyzer as for operating mode - pay careful attention to gas leak tightness.

GMS800

9 Clearing Malfunctions

General malfunctions
Malfunction messages
Display messages
Measurement errors

9.1 If the GMS800 does not work at all ...

Possible causes	Information
Mains cable not connected.	► Check mains cable and connections.
Main switch off.	► Check main switch (external).[1]
Mains supply shut off.	► Check mains supply (e.g. socket, external fuses).
<i>On enclosures with enclosure pressurization system:</i> The enclosure pressurization system has automatically interrupted the mains supply (safety switch-off).	► Check the status of the enclosure pressurization system.
Internal mains fuse defective.	► Have internal fuse checked (information → p. 26, § 3.4.2).
Internal operating temperatures not correct.	► Check whether corresponding malfunction message displays are present.
Sample gas feed not working.	► Check (→ p. 23, § 3.3.2).
Internal software not working.	Can only occur after complex internal malfunctions or after strong external interference (e.g. strong electromagnetic interference impulses). ► Switch off the GMS800. Wait for a few seconds, then switch on again.
Internal overheat fuse triggered	Heated Analyzer modules have overheating fuses which are defective after triggering. ► Inform the manufacturer's Customer Service to have the defective overheating fuse replaced.

[1] The GMS800 has no own mains switch.

9.2 Malfunction displays

The control unit activates the malfunction field when a module signals an internal malfunction (→ Operating Instructions for Control Unit).

- *To localize the malfunction cause:* Call up the Diagnosis → State function in the menu branch for each module and check whether one of the LED symbols for “Failure”, “Maintenance” or “Unsafe State” is activated.

If this is the case:

- Call up the Diagnosis function “Logbook” and check the current entries.
- Inform the skilled person trained and authorized to clear malfunctions or contact the manufacturer's Customer Service.



- The Logbook contains the malfunctions of a module in tabular form with an error code (→ Supplementary Operating Instructions for Modules).
- “SOPAS ET” blends in the significance of the error code after clicking the Logbook table once.

9.3

If measured values are obviously incorrect ...

Possible causes	Information	Notes for service
GMS800 not ready for operation.	► Check operational readiness (→ p. 35, § 5.3).	–
GMS800 not measuring the sample gas.	► Check sample gas path and all valves (e.g. switching from test gas to sample gas).	► Make sure valves are functioning correctly, disassemble if necessary.
Sample gas path not switched correctly.		
GMS800 not correctly adjusted.	► Check prerequisites for correct adjustment: – Correct test gases used? – Nominal values set correctly? ► Then run an adjustment.	► Check test gases used critically (nominal values, manufacturing tolerance, state).
Measuring parameter settings not suitable for the application.	► Check corresponding settings (e.g. damping). Try a different setting as test.	–
Sample gas pressure inside the GMS800 too high.	► Ensure sample gas pressure in the GMS800 is not greater than 20 kPa (= 200 mbar) against atmospheric pressure.	The gas pressure can influence measured values in most of the measuring principles used.
Sample gas path not gas-tight.	► Visually inspect the installation. ► <i>When a defect is suspected:</i> Inform the manufacturer's Customer Service or trained skilled persons.	Carry out a leak tightness check.
When only observed on one measured value output: The load is too high.	► Make sure that the total internal resistance of the connected devices is not larger than 500 Ω.	► Measure including feed line.
Analyzer module is contaminated.	► Contact manufacturer's Customer Service or trained skilled persons.	► Inspect the measuring cell/cuvette. ► Clean or replace if necessary.
With computation of an analog input (option): External analog signal is erroneous or has failed.	► Check external equipment that delivers the analog signal for cross-sensitivity compensation.	– Connection interrupted? – Problem with external measurement? – External analyzer not adjusted?

9.4

If measured values are unstable for no apparent reason ...

Possible causes	Information	Notes for service
Strong pressure fluctuations at the sample gas outlet.	► Install separate exhaust gas line for the GMS800.	–
Strong mechanical vibrations.	► Check ambient conditions where the GMS800 is installed.	–

GMS800

10 Technical Data (Information)

The Technical Data are specified in the following documents:

Technical data for	See
Enclosure specifications	→ Supplementary Operating Instructions for Enclosure.
Ambient conditions, climate	
Gas connection details	
Mains connection	
Electrical safety	
Technical gas requirements	→ Supplementary Operating Instructions for the Analyzer Modules fitted
Metrological data	→ Supplementary Operating Instructions for "I/O Module"
Signal connections	

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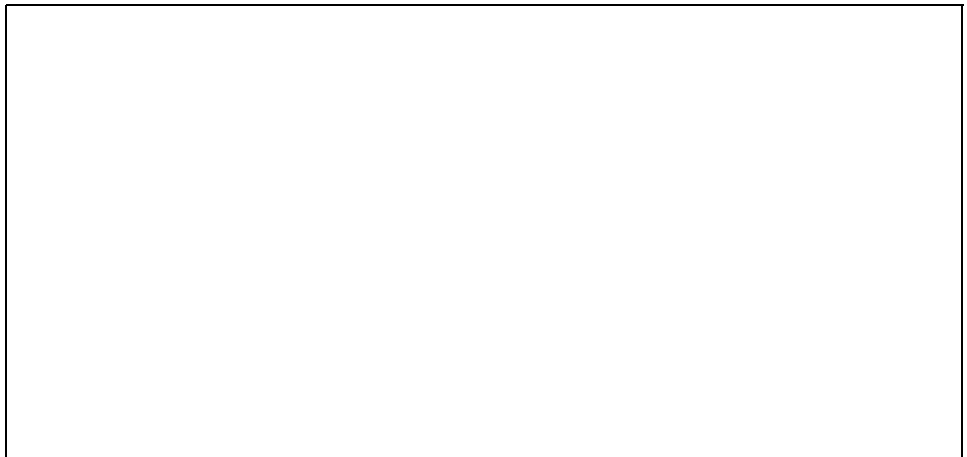
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